

Harvestify Project Architecture & File Structure

1. Backend Files (Server & Logic)

app/app.py	Main backend Flask server handling routes, API requests, and model inference.
app/config.py	Backend configuration settings and constants.
app/utils/disease.py	Backend utility for mapping plant diseases from the computer vision model.
app/utils/fertilizer.py	Backend utility mapping fertilizer recommendations.
app/utils/model.py	Backend script containing the ResNet9 PyTorch architecture.
train_irrigation.py	Backend script to train the custom irrigation ML model.
requirements.txt	Lists all Python packages needed to run the backend.
.env	Stores hidden backend environment variables like API keys.

2. Frontend Files (User Interface)

app/templates/*.html	Frontend views (HTML). Examples: index.html (Home), chatbot.html, disease.html.
app/templates/layout.html	Frontend master template containing the navbar and the newly designed footer.
app/static/css/style.css	Frontend styling containing custom CSS rules for the UI.
app/static/css/bootstrap.css	Frontend styling library for layout structure.
app/static/scripts/cities.js	Frontend Javascript file handling dynamic state/city dropdowns.
app/static/data/fallback_mandi_prices.json	Frontend JSON data used when the Mandi API fails.
app/static/images/*	Frontend image assets like logo.png, backgrounds, and favicons.

3. Machine Learning Models (Pre-trained)

models/RandomForest.pkl	ML model used for crop recommendation.
models/fertilizer_model.pkl	ML model used for predicting fertilizer requirements.
models/plant-disease-model-new.pth	Deep Learning PyTorch model for plant disease classification.
models/production_pipeline.pkl	ML pipeline for predicting crop yield/production.
models/irrigation_model.pkl	ML model for estimating water requirements for irrigation.

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4. Data Science Notebooks (Jupyter)

notebooks/Crop_Recommendation_Model.ipynb	Notebook for training the crop recommendation ML model.
notebooks/fertilizer.ipynb	Notebook exploring and training the fertilizer prediction model.
notebooks/plant-disease-classification.ipynb	Notebook demonstrating ResNet9 training on plant leaves.
notebooks/Yield Prediction.ipynb	Notebook analyzing crop yield factors.

5. Datasets (CSV & XLS)

Data-raw/*.csv	Raw, unprocessed datasets (e.g., crop_yield.xls, cpdata.csv) downloaded from sources.
Data-processed/*.csv	Cleaned and processed datasets (e.g., crop_recommendation.csv) used directly for ML training.